



DIGITAL VLSI DESIGN

Prof. Mahesh Awati & Prof. Vinay Reddy Narayana
Department of Electronics and Communication Engg.

DIGITAL VLSI DESIGN

PART 1

MOS Inverters (Static Characteristics)

Vinay Reddy Narayana

Department of Electronics and Communication Engineering

MOS Inverters (Static Characteristics)

UNIT 1 (Part 1) : MOS Inverters (Static Characteristics)

VLSI design Methodologies and Design Flow (1.4 &1.5)

Introduction to inverters (5.1)

Resistive-Load Inverter (5.2)

Inverters with n-Type MOSFET Load (5.3)

CMOS Inverter (5.4)

DIGITAL VLSI DESIGN

MOS Inverters (Static Characteristics)

Part c - Resistive Load Inverter

Vinay Reddy Narayana

Department of Electronics and Communication Engineering

MOS Inverters (Static Characteristics)

Example 5.1

Consider the following inverter design problem: Given $V_{DD} = 5 \text{ V}$, $k_n' = 30 \mu\text{A/V}^2$, and $V_{T0} = 1 \text{ V}$, design a resistive-load inverter circuit with $V_{OL} = 0.2 \text{ V}$. Specifically, determine the (W/L) ratio of the driver transistor and the value of the load resistor R_L that achieve the required V_{OL} .

$$\frac{(V_{DD} - V_{OL})}{R_L} = \frac{k_n'}{2} \left(\frac{W}{L} \right) [2(V_{in} - V_{T0})V_{OL} - V_{OL}^2]$$

$$\frac{(V_{DD} - V_{OL})}{R_L} = \frac{k_n'}{2} \left(\frac{W}{L} \right) [2(V_{OH} - V_{T0})V_{OL} - V_{OL}^2]$$

$$\frac{(5 - 0.2)}{R_L} = \frac{30 \times 10^{-6}}{2} \left(\frac{W}{L} \right) [2(5 - 1)0.2 - 0.2^2]$$

$$\left(\frac{W}{L} \right) R_L = 2.05 \times 10^5$$

MOS Inverters (Static Characteristics)

$\left(\frac{W}{L}\right)$ - Ratio	Load Resistor R_L [k Ω]	DC Power Consumption $P_{DC,average}$ [μ W]
1	205.0	58.5
2	102.5	117.1
3	68.4	175.4
4	51.3	233.9
5	41.0	292.7
6	34.2	350.8

Power consumption increases as the value of the load resistor R_L is decreases and (W/L) increases.

If Power consumption is the concern, (W/L) is kept LOW and a large load resistor (R_L) need to be considered but large resistor will require more area of silicon for fabrication.

Example 5.2

Consider a resistive-load inverter circuit with $V_{DD} = 5\text{ V}$, $k_n' = 20\text{ }\mu\text{A/V}^2$, $V_{TD} = 0.8\text{ V}$, $R_L = 200\text{ k}\Omega$, and $W/L = 2$. Calculate the critical voltages (V_{OL} , V_{OH} , V_{IL} , V_{IH}) on the VTC and find the noise margins of the circuit.

$$V_{OL} = 0.147\text{ V}, V_{OH} = 5\text{ V}, V_{IL} = 0.925\text{ V}, V_{IH} = 1.97\text{ V}, \\ NM_L = 0.78\text{ V}, NM_H = 3.03$$

Observations: NML is quite Low and which may lead to misinterpretation of input signal level.

For better noise immunity, the noise margin for 'LOW' signal should be at least 25% of VDD i.e 1.25V for VDD=5V

Exercise Problem 5.1 – Similar to Example 5.1

5.1 Design a resistive-load inverter with $R = 1 \text{ k}\Omega$, such that $V_{OL} = 0.6 \text{ V}$.

The enhancement-type nMOS driver transistor has the following parameters:

$$V_{DD} = 5.0 \text{ V}$$

$$V_{T0} = 1.0 \text{ V}$$

$$\gamma = 0.2 \text{ V}^{1/2}$$

$$\lambda = 0$$

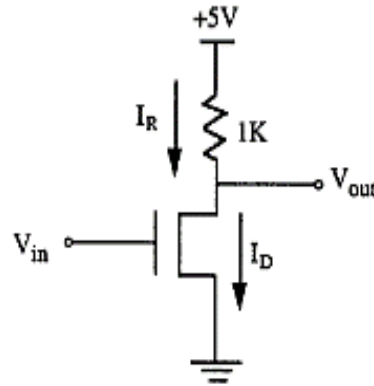
$$\mu_n C_{ox} = 22.0 \text{ }\mu\text{A/V}^2$$

- (a) Determine the required aspect ratio, W/L .
- (b) Determine V_{IL} and V_{IH} .
- (c) Determine noise margins NM_L and NM_H .

MOS Inverters (Static Characteristics)

SOLUTION:

(a)



When $V_{out} = V_{OL}$, $V_{in} = V_{DD} = 5.0V$, the driver transistor operates in linear region. Using Eq.(5.12)

$$I_R = \frac{V_{DD} - V_{OL}}{1K} = \frac{k'}{2} \frac{W}{L} [2(V_{OH} - V_{T0})V_{OL} - V_{OL}^2]$$

$$\frac{5 - 0.6}{1000} = \frac{22 \times 10^{-6}}{2} \frac{W}{L} [2(5 - 1)0.6 - 0.6^2]$$

Solve for W/L ,

$$\frac{W}{L} = 90.1$$

$$V_{IL} = V_{T0} + \frac{1}{R_L \cdot k' \frac{W}{L}} = 1.0 + \frac{1}{1000 \cdot 22 \times 10^{-6} \cdot 90.1} = 1.5[V]$$

$$V_{IH} = V_{T0} + \sqrt{\frac{8}{3} \cdot \frac{V_{DD}}{R_L \cdot k' \frac{W}{L}} - \frac{1}{R_L \cdot k' \frac{W}{L}}} \\ = 1.0 + \sqrt{\frac{8}{3} \frac{5}{1000 \cdot 22 \times 10^{-6} \cdot 90.1} - \frac{1}{1000 \cdot 22 \times 10^{-6} \cdot 90.1}} = 3.09[V]$$

$$NM_L = V_{IL} - V_{OL} = 1.5 - 0.6 = 0.9[V]$$

$$NM_H = V_{OH} - V_{IH} = 5.0 - 3.09 = 1.91[V]$$

DIGITAL VLSI DESIGN

MOS Inverters (Static Characteristics)

Part e - CMOS Inverter

Vinay Reddy Narayana

Department of Electronics and Communication Engineering

MOS Inverters (Static Characteristics)

Example 5.4

Consider a CMOS inverter circuit with the following parameters :

$$V_{DD} = 3.3 \text{ V}$$

$$V_{T0,n} = 0.6 \text{ V}$$

$$V_{T0,p} = -0.7 \text{ V}$$

$$k_n = 200 \mu\text{A/V}^2$$

$$k_p = 80 \mu\text{A/V}^2$$

Calculate the noise margins of the circuit. Notice that the CMOS inverter being considered here has $k_R = 2.5$ and $V_{T0,n} \neq |V_{T0,p}|$; hence, it is not a symmetric inverter.

MOS Inverters (Static Characteristics)

First, the output low voltage V_{OL} and the output high voltage V_{OH} are found, using (5.54) and (5.55), as $V_{OL} = 0$ and $V_{OH} = 5$ V. To calculate V_{IL} in terms of the output voltage, we use (5.62).

$$V_{IL} = \frac{2V_{out} + V_{T0,p} - V_{DD} + k_R V_{T0,n}}{1 + k_R}$$

$$= \frac{2V_{out} - 0.7 - 3.3 + 1.5}{1 + 2.5} = 0.57V_{out} - 0.71$$

Now substitute this expression into the KCL equation (5.59).

$$\frac{k_n}{2} \cdot (V_{in} - V_{T0,n})^2 = \frac{k_p}{2} \cdot \left[2 \cdot (V_{in} - V_{DD} - V_{T0,p}) \cdot (V_{out} - V_{DD}) - (V_{out} - V_{DD})^2 \right] \quad (5.59)$$

MOS Inverters (Static Characteristics)

$$2.5(0.57 V_{out} - 0.71 - 0.6)^2 = 2(0.57 V_{out} - 0.71 - 3.3 + 0.7)(V_{out} - 3.3) - (V_{out} - 3.3)^2$$

This expression yields a second-order polynomial in V_{out} , as follows:

$$0.66 V_{out}^2 + 0.05 V_{out} - 6.65 = 0$$

Only one root of this quadratic equation corresponds to a physically correct solution for V_{out} (i.e., $V_{out} > 0$).

$$V_{out} = 3.14 \text{ V}$$

From this value, we can calculate the critical voltage V_{IL} as:

$$V_{IL} = 0.57 \cdot 3.14 - 0.71 = \underline{\underline{1.08 \text{ V}}}$$

To calculate V_{IH} in terms of the output voltage, use (5.67):

$$\begin{aligned} V_{IH} &= \frac{V_{DD} + V_{T0,p} + k_R \cdot (2 V_{out} + V_{T0,n})}{1 + k_R} \\ &= \frac{3.3 - 0.7 + 2.5(2 V_{out} + 0.6)}{1 + 2.5} = 1.43 V_{out} + 1.17 \end{aligned}$$

Next, substitute this expression into the KCL equation (5.64) to obtain a second-order polynomial in V_{out} .

$$2.5[2(1.43 V_{out} + 1.17 - 0.6) V_{out} - V_{out}^2] = (1.43 V_{out} - 1.43)^2$$

$$2.61 V_{out}^2 + 6.94 V_{out} - 2.04 = 0$$

Again, only one root of this quadratic equation corresponds to the physically correct solution for V_{out} at this operating point, i.e., when $V_{in} = V_{IH}$.

$$V_{out} = 0.27 \text{ V}$$

From this value, we can calculate the critical voltage V_{IH} as:

$$V_{IH} = 1.43 \cdot 0.37 + 1.17 = \underline{\underline{1.55 \text{ V}}}$$

MOS Inverters (Static Characteristics)

Finally, we find the noise margins for low voltage levels and for high voltage levels using (5.3) and (5.4).

$$NM_L = V_{IL} - V_{OL} = 1.08 \text{ V}$$

$$NM_H = V_{OH} - V_{IH} = 1.75 \text{ V}$$

MOS Inverters (Static Characteristics)

5.6 Consider a CMOS inverter with the following parameters:

<i>nMOS</i>	$V_{T0,n} = 0.6 \text{ V}$	$\mu_n C_{ox} = 60 \mu\text{A/V}^2$	$(W/L)_n = 8$
<i>pMOS</i>	$V_{T0,p} = -0.7 \text{ V}$	$\mu_p C_{ox} = 25 \mu\text{A/V}^2$	$(W/L)_p = 12$

Calculate the noise margins and the switching threshold (V_{th}) of this circuit. The power supply voltage is $V_{DD} = 3.3 \text{ V}$.

MOS Inverters (Static Characteristics)

5.6 Consider a CMOS inverter with the following parameters:

<i>nMOS</i>	$V_{T0,n} = 0.6 \text{ V}$	$\mu_n C_{ox} = 60 \mu\text{A/V}^2$	$(W/L)_n = 8$
<i>pMOS</i>	$V_{T0,p} = -0.7 \text{ V}$	$\mu_p C_{ox} = 25 \mu\text{A/V}^2$	$(W/L)_p = 12$

Calculate the noise margins and the switching threshold (V_{th}) of this circuit. The power supply voltage is $V_{DD} = 3.3 \text{ V}$.



THANK YOU

Vinay Reddy

Department of Electronics and Communication

vinay@pes.edu

+91 9945730244